

The Min Cost Flow Problem

– *augmenting cycle method*

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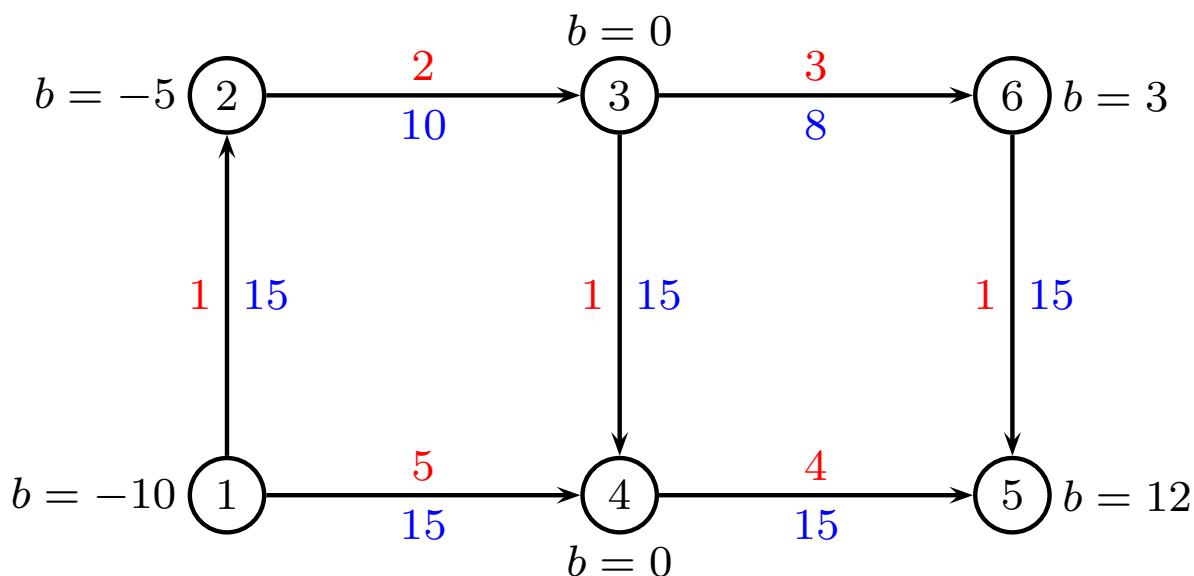
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Min Cost Flow - Terminology

- We consider a directed graph $G = (V, E)$ in which each edge e has a capacity $u_e \in \mathbb{R}_+$ and a unit transportation cost $w_e \in \mathbb{R}$.
- Each vertex i furthermore has a demand $b_i \in \mathbb{R}$. If $b_i \geq 0$ then i is called a **sink**, and if $b_i < 0$ then i is called a **source**.
- We assume that $b_V = \sum_{i \in V} b_i = 0$.



Min Cost Flow - Definition

The Min Cost Flow problem consists in supplying the sinks from the sources by a flow in the cheapest possible way:

$$\begin{array}{ll}\min & \sum_{(i,j) \in E} w_{ij} x_{ij} \\ \text{s.t.} & f_x(i) = b_i \quad i \in V \\ & 0 \leq x_{ij} \leq u_{ij} \quad (i,j) \in E\end{array}$$

where $f_x(i) = \sum_{(j,i) \in E} x_{ji} - \sum_{(i,j) \in E} x_{ij}$.

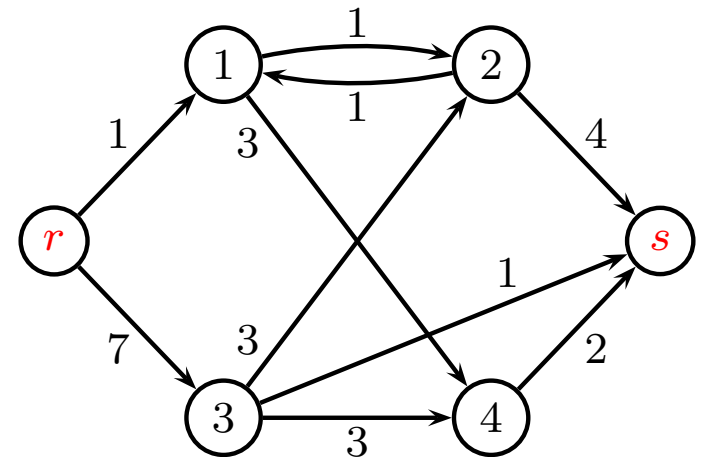
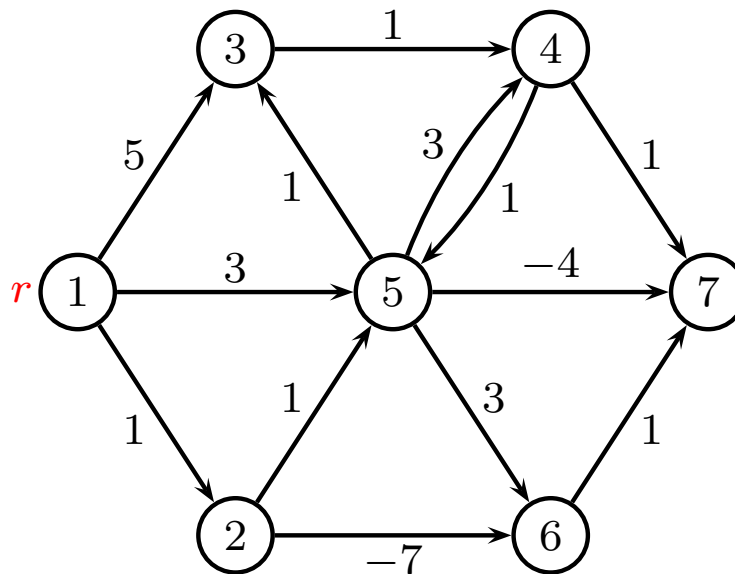
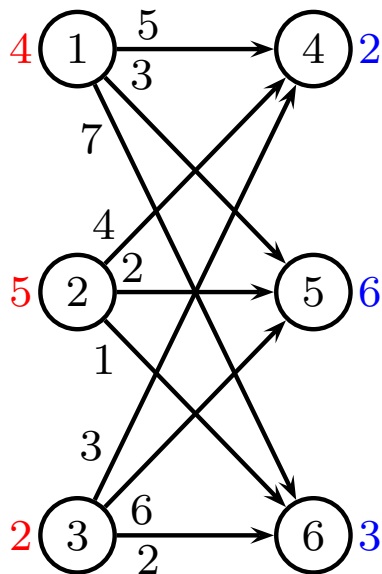
- Flow balance constraints
- Capacity constraints

Often integral flow: $x_{ij} \in \mathbb{Z}$

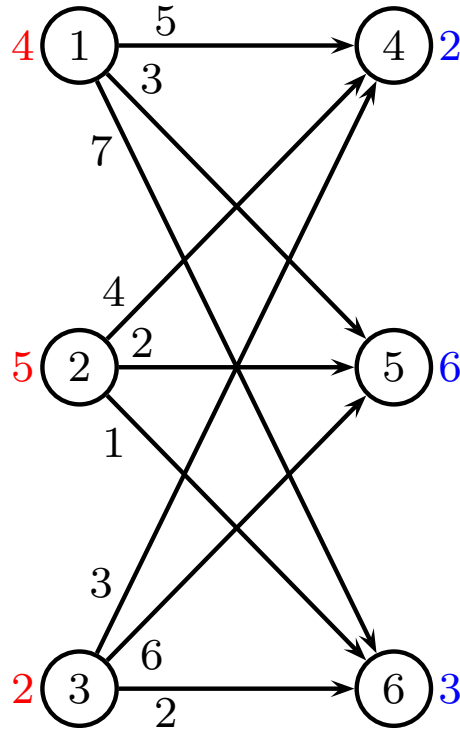
Special cases

Numerous flow problems can be stated as a Min Cost Flow problem:

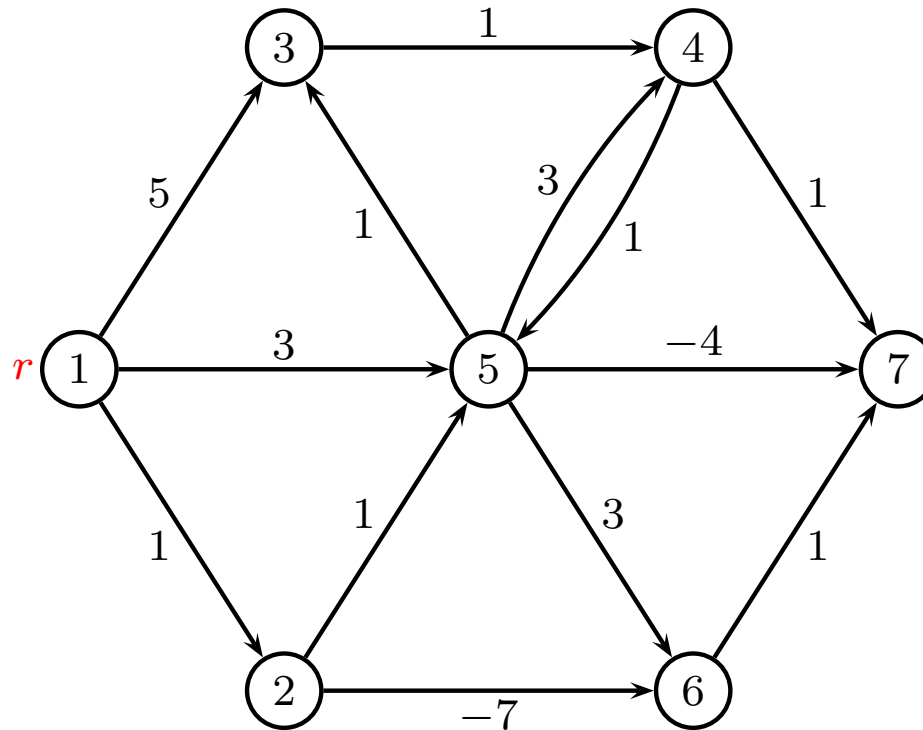
1. The Transportation Problem
2. The Shortest Path Problem
3. The Max Flow Problem



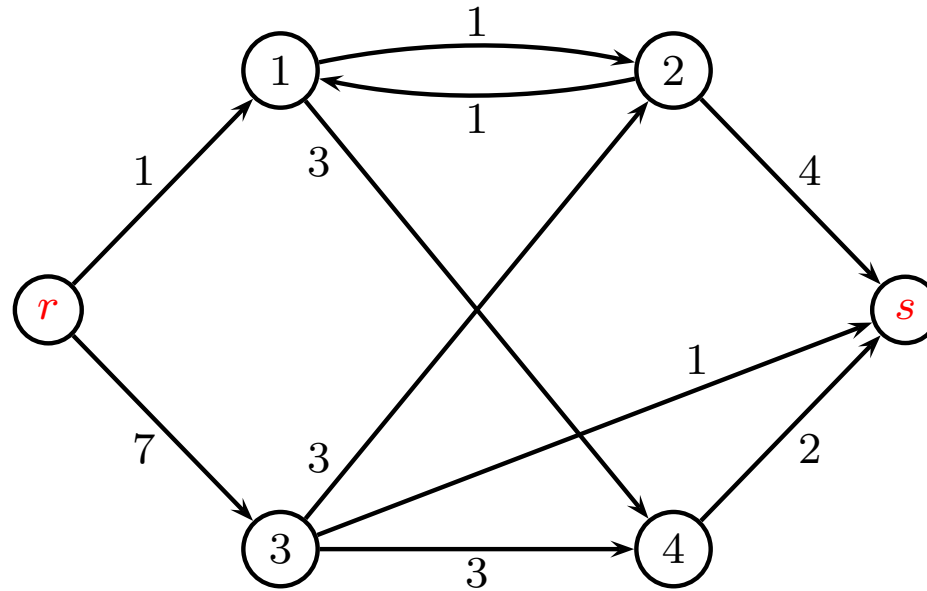
Transportation Problem



Shortest Path Problem



Max Flow Problem



TU and Min Cost Flow

- **(Prop. 3.1):** A matrix A is TU if and only if (i) the transpose matrix A^T is TU if and only if (ii) the matrix (A, I) is TU.
- **(Prop. 3.2):** (Sufficient condition). A matrix A is TU if
 1. $a_{ij} \in \{+1, -1, 0\}$ for all i, j .
 2. Each column contains at most two nonzero coefficients ($\sum_{i=1}^m |a_{ij}| \leq 2$).
 3. There exists a partition (M_1, M_2) of the set M of rows such that each column j containing two nonzeros satisfies $\sum_{i \in M_1} a_{ij} - \sum_{i \in M_2} a_{ij} = 0$.

TU and Min Cost Flow

	x_{e_1}	x_{e_2}	...	x_{ij}	...	x_{e_m}		
	w_{e_1}	w_{e_2}	...	w_{ij}	...	w_{e_m}		
1	-1	.	...		...		=	b_1
2	.	-1	...	.	...	.	=	b_2
.	.	.	...	.	...	.	=	.
i	1	1	...	-1	...	1	=	b_i (y_i)
.	.	.	...	.	...	.	=	
j	.	.	...	1	...	.	=	b_j (y_j)
.	.	.	...	.	...	-1	=	.
n	.	.	...	.	...	.	=	b_n
e_1	-1						$\geq$	$-u_1$
e_2		-1					$\geq$	$-u_2$
.	.	.	.	.	.	.	$\geq$	.
(i, j)				-1			$\geq$	$-u_{ij}$ (z_{ij})
.	.	.	.	.	.	.	$\geq$	.
e_m						-1	$\geq$	$-u_m$

Min Cost Flow - Dual LP

$$\begin{array}{ll}\min & \sum_{(i,j) \in E} w_{ij} x_{ij} \\ \text{s.t.} & \sum_{(j,i) \in E} x_{ji} - \sum_{(i,j) \in E} x_{ij} = b_i \quad i \in V \quad (y_i) \\ & -x_{ij} \geq -u_{ij} \quad (i,j) \in E \quad (z_{ij}) \\ & x_{ij} \geq 0 \quad (i,j) \in E\end{array}$$

Dual problem:

$$\begin{array}{ll}\max & \sum_{i \in V} b_i y_i - \sum_{(i,j) \in E} u_{ij} z_{ij} \\ \text{s.t.} & -w_{ij} - y_i + y_j \leq z_{ij} \quad (i,j) \in E \quad (x_{ij}) \\ & z_{ij} \geq 0 \quad (i,j) \in E\end{array}$$

Reduced cost $\bar{w}_{ij} = w_{ij} + y_i - y_j$. Last constraint becomes

$$-\bar{w}_{ij} \leq z_{ij} \quad (i,j) \in E \quad (x_{ij})$$

Optimality Conditions I

- If $u_{ij} = +\infty$ (i.e. no capacity constraints for e) then z_{ij} must be 0 and hence $\bar{w}_{ij} \geq 0$ just has to hold (primal optimality condition for the LP).
- If $u_{ij} \neq +\infty$ then $z_{ij} \geq 0$ and $z_{ij} \geq -\bar{w}_{ij}$ must hold. z has **negative** coefficient in the objective function – hence the best choice for z is as small as possible:

$$z_{ij} = \max\{0, -\bar{w}_{ij}\}$$

- Therefore, the optimal value of z_{ij} is uniquely determined from the other variables, and z_{ij} is “unnecessary” in the dual problem.

Optimality Conditions I - Comp. Slackness

Complementary slackness conditions

$$\begin{array}{ll} -\bar{w}_{ij} = z_{ij} & \vee \quad x_{ij} = 0 \\ x_{ij} = u_{ij} & \vee \quad z_{ij} = 0 \end{array}$$

This means

$$\begin{aligned} x_{ij} > 0 &\Rightarrow -\bar{w}_{ij} = z_{ij} = \max(0, -\bar{w}_{ij}) \\ \text{i.e. } (x_{ij} > 0 &\Rightarrow -\bar{w}_{ij} \geq 0) \end{aligned}$$

and

$$\begin{aligned} z_{ij} > 0 &\Rightarrow x_{ij} = u_{ij} \\ \text{i.e. } (-\bar{w}_{ij} > 0 &\Rightarrow x_{ij} = u_{ij}) \end{aligned}$$

$$\begin{array}{ll} x_{ij} > 0 & \Rightarrow \quad \bar{w}_{ij} \leq 0 \\ x_{ij} < u_{ij} & \Rightarrow \quad \bar{w}_{ij} \geq 0 \end{array}$$

Residual Graph I

Idea:

- Find initial feasible flow
 - Consider residual graph
 - Find augmenting path (cycle) in residual graph
 - Construct new feasible flow
-
- For a legal flow x in G , the residual graph is (like for Max Flow) a graph, in which the paths indicate **how flow excess can be moved** in G given that the flow x already is present. The only difference is that each edge has a cost assigned.

Residual Graph II

- The residual graph G_x for G wrt. x is defined by

$$\begin{aligned} V(G_x) &= V \\ E(G_x) &= E_x = \{(i, j) : (i, j) \in E \wedge x_{ij} < u_{ij}\} \\ &\quad \cup \{(i, j) : (j, i) \in E \wedge x_{ji} > 0\} \end{aligned}$$

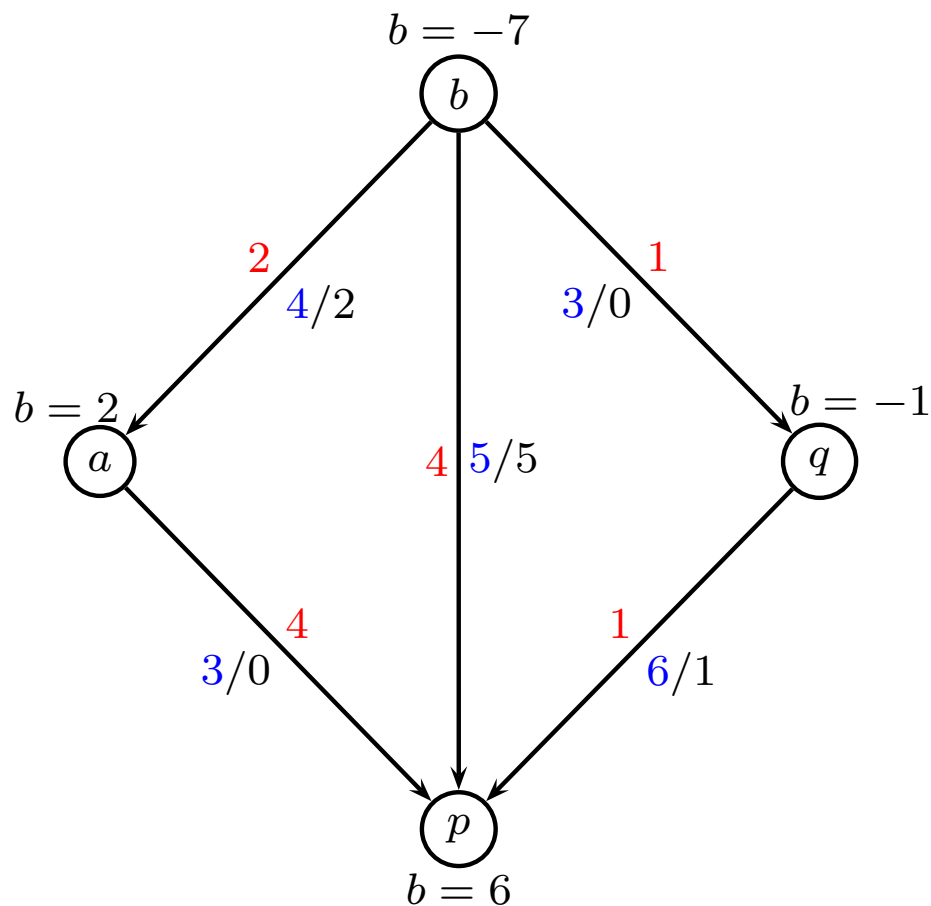
- The unit cost

$$w'_{ij} = \begin{cases} w_{ij} & \text{if } x_{ij} < u_{ij} \\ -w_{ji} & \text{if } x_{ji} > 0 \end{cases}$$

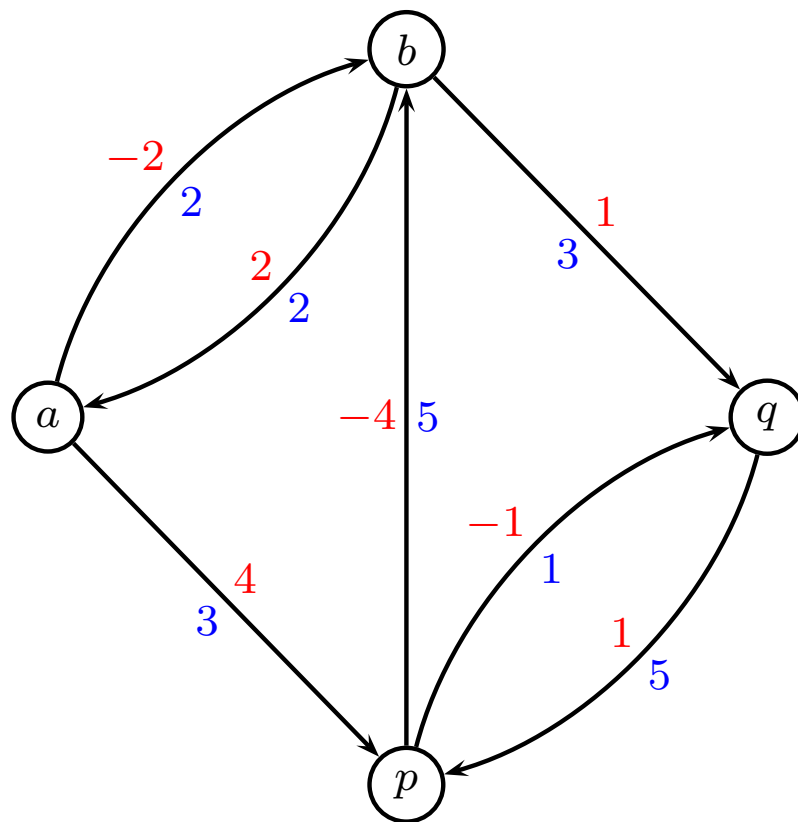
- The residual capacity

$$r_{ij} = \begin{cases} u_{ij} - x_{ij} & \text{if } x_{ij} < u_{ij} \\ x_{ji} & \text{if } x_{ji} > 0 \end{cases}$$

Residual Graph



Graph with flow



Residual graph with costs and capacities

Residual Graph III

- Note that a directed circuit with negative cost in G_x corresponds to a negative cost circuit in G , if cost are added for forward edges and subtracted for backward edges.
- Note that if a set of potentials $y_i, i \in V$ are given, and the cost of a circuit wrt. the reduced costs for the edges ($\bar{w}_{ij} = w_{ij} + y_i - y_j$) are calculated, the cost remains the same as the original costs – the potentials are “telescoped” to 0.

$$\sum_C \bar{w}_{ij} = \sum_C w_{ij} + \sum_C y_i - \sum_C y_j = \sum_C w_{ij}$$

- We normally use reduced costs to drive search (greedy principle) but same as using original costs

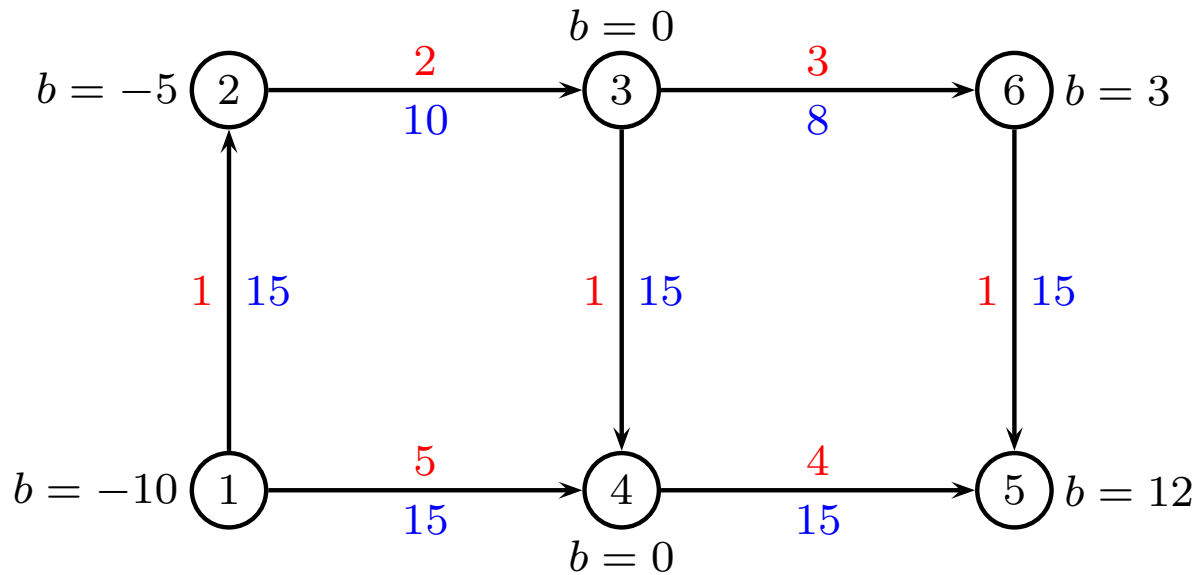
Min Cost Flow - Negative cost circuits

- A primal feasible flow satisfying sink demands from sources and respecting the capacity constraints is **optimal if and only if** an x -augmenting circuit with negative w -cost (or negative $\overline{w}$ -cost – there is no difference) does not exist.
(will be proved later)
- This is the idea behind the identification of optimal solutions in the **network simplex algorithm**.

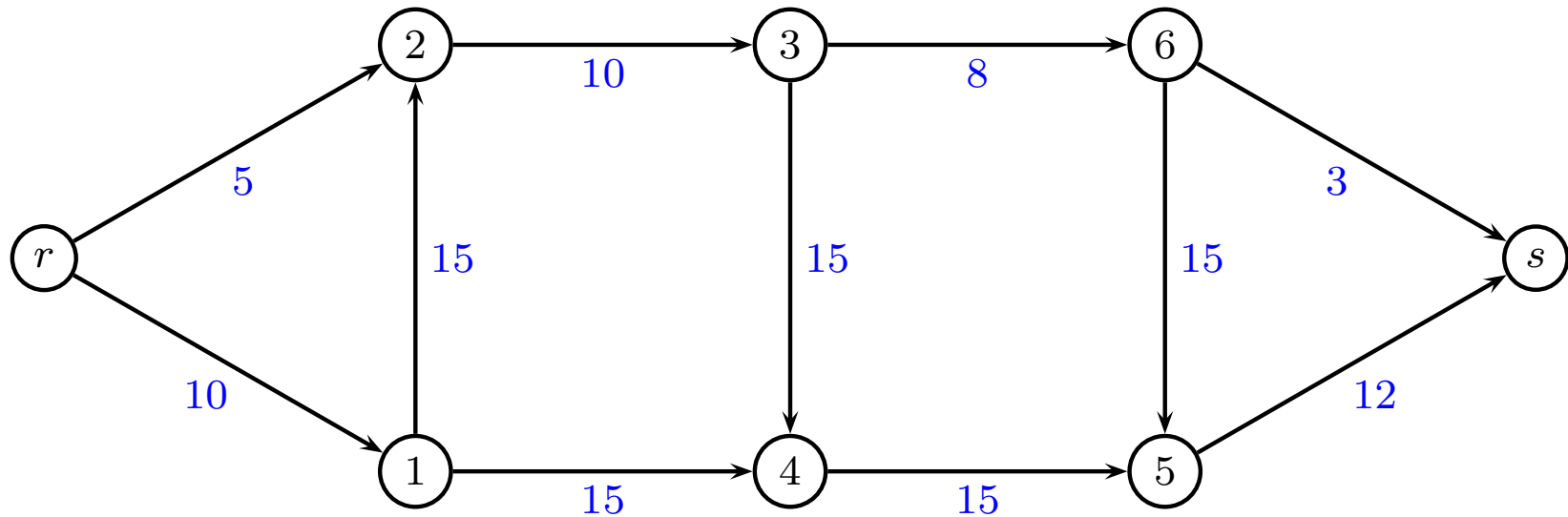
Augmenting Circuit Algorithm

- 1 find a feasible solution x ;
 - 2 **while** *there exists an augmenting circuit* **do**
 - 3 find an augmenting circuit C ;
 - 4 find “bottleneck” as minimum residual capacity in C
$$\theta = \min_{(i,j) \in C} \{r_{ij}\} ;$$
 - 5 augment x on C with θ units ;
- initial flow is found by solving max-flow problem
 - negative circuit is found by Bellman-Ford in $O(nm)$

Example

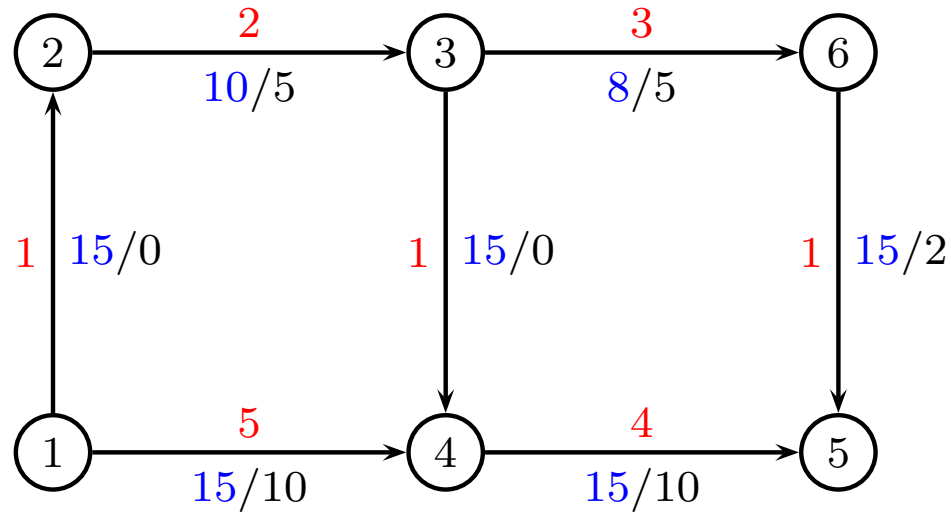


Initialize



Drop demands in the following

Example



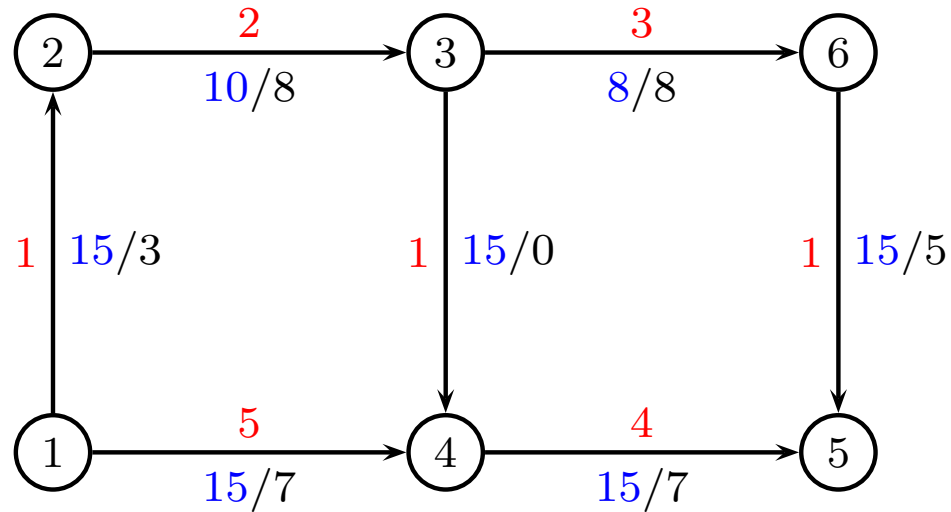
cost: $z = 2 \cdot 5 + 3 \cdot 5 + 1 \cdot 2 + 5 \cdot 10 + 4 \cdot 10 = 117$

cycle: $C = \{2, 3, 6, 5, 4, 1\}$

$w(C) = 2 + 3 + 1 - 4 - 5 + 1 = -2$

$\theta = \min\{5, 3, 13, 10, 10, 15\} = 3$

Example



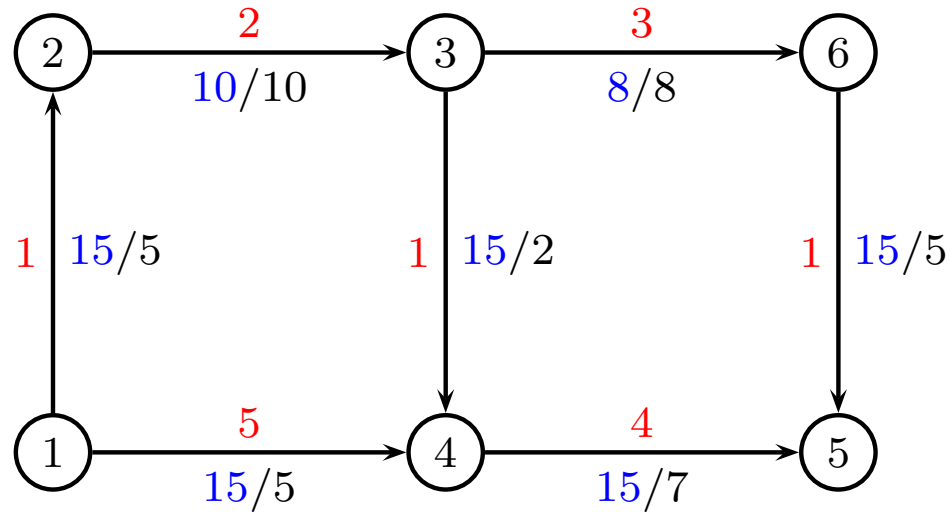
cost: $z = 2 \cdot 8 + 3 \cdot 8 + 1 \cdot 3 + 1 \cdot 5 + 5 \cdot 7 + 4 \cdot 7 = 111$

cycle: $C = \{2, 3, 4, 1\}$

$w(C) = 2 + 1 - 5 + 1 = -1$

$\theta = \min\{2, 15, 7, 12\} = 2$

Example



cost: $z = 2 \cdot 10 + 3 \cdot 8 + 1 \cdot 5 + 1 \cdot 2 + 1 \cdot 5 + 5 \cdot 5 + 4 \cdot 7 = 109$

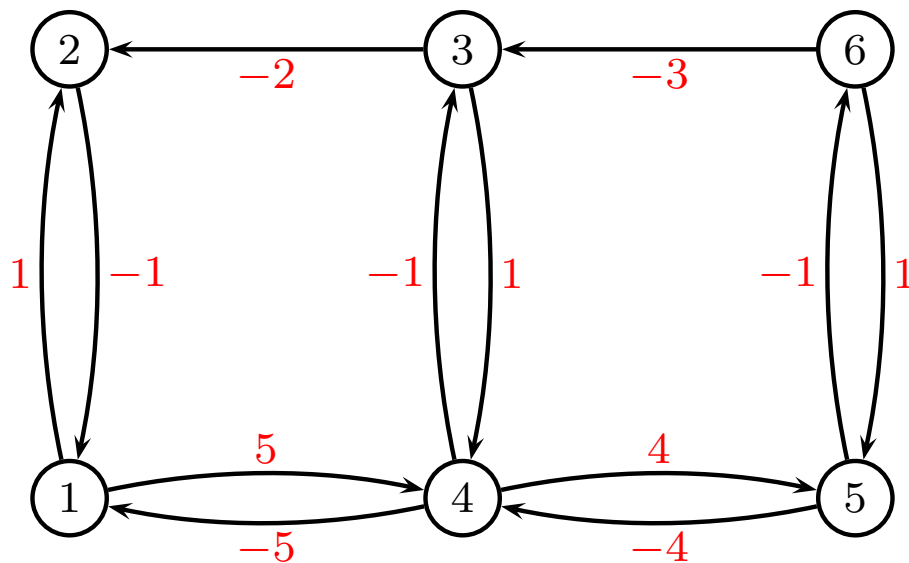
No negative cycles

stop

Proof of termination criteria

A flow x is optimal $\Leftrightarrow$ no augmenting circuit can be found

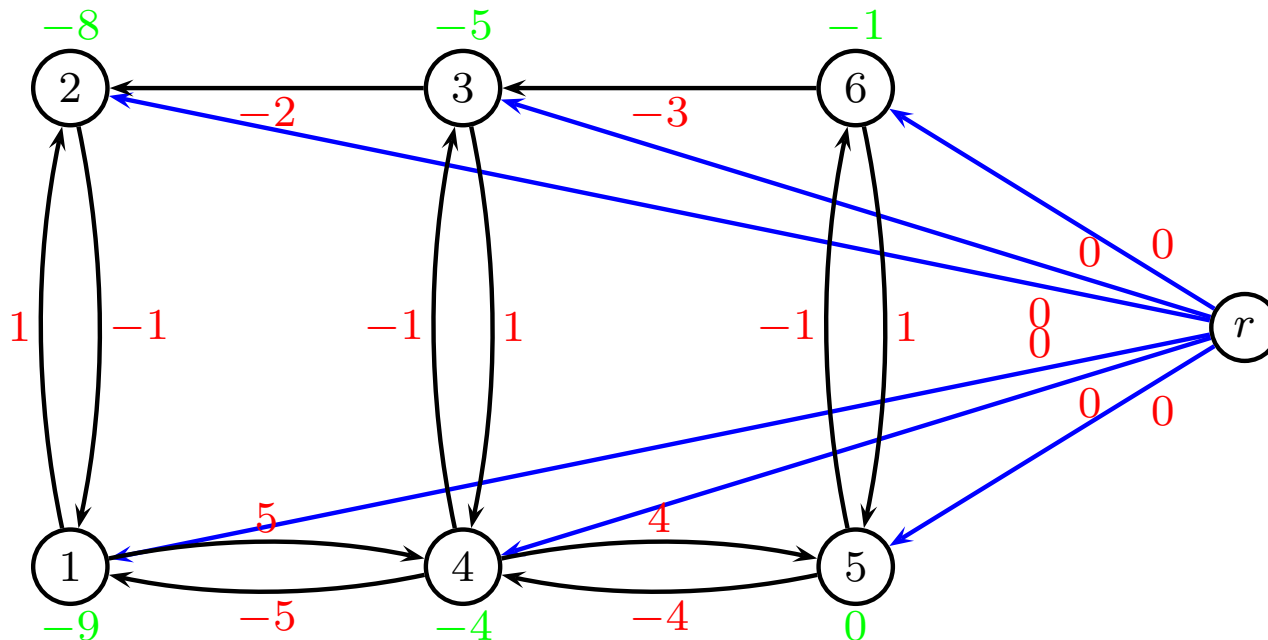
- Residual graph (only costs):



○

- if negative circuit can be found then construct better solution
- now, assume no negative circuit can be found in residual graph

Proof of termination criteria



Shortest path from r to i defines y_i . Satisfies $y_i + w'_{ij} \geq y_j$

$$y_i + w_{ij} \geq y_j \quad \text{if } x_{ij} < u_{ij}$$

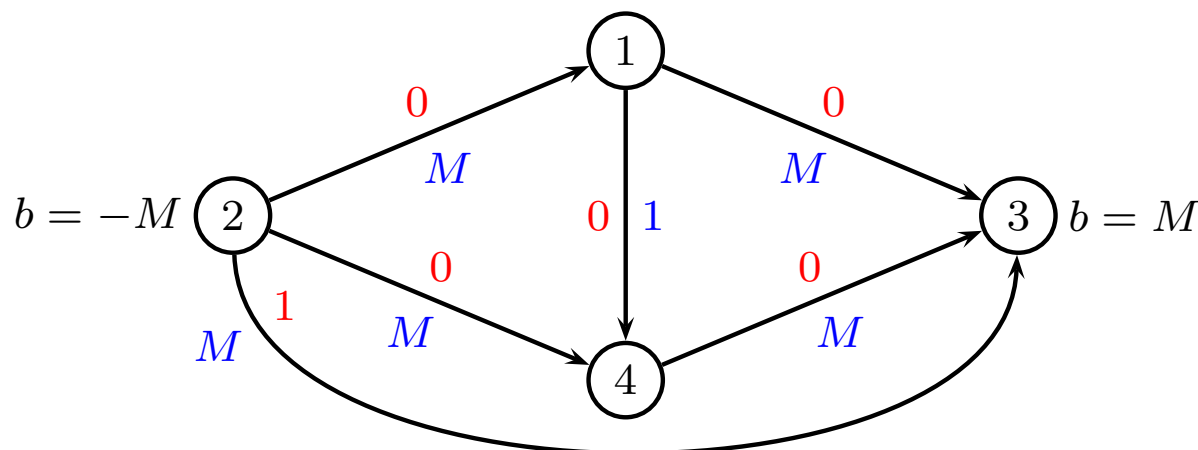
$$y_i - w_{ji} \geq y_j \quad \text{if } x_{ji} > 0$$

since $\bar{w}_{ij} = w_{ij} + y_i - y_j$ we have

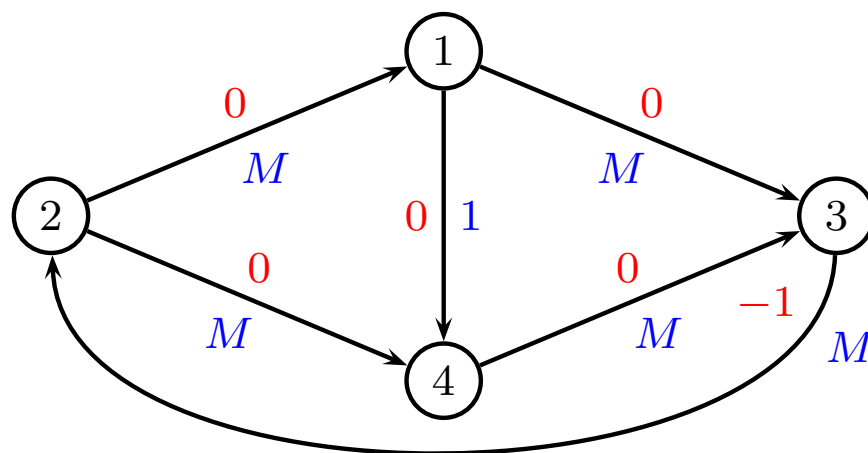
$$\bar{w}_{ij} \geq 0 \quad \text{if } x_{ij} < u_{ij}$$

$$\bar{w}_{ji} \leq 0 \quad \text{if } x_{ji} > 0$$

Time Complexity



Initialize: send M units along bottom edge. Residual graph



Negative cycle $C = \{2, 1, 4, 3\}$ then $C = \{2, 4, 1, 3\}$ then ...
(Similar to proof for max-flow augmenting path method)

Time Complexity

- Finding negative cycle takes $O(nm)$ with Bellman-Ford
- Objective $z \leq mWU$ where $w_{ij} \leq W$ and $u_{ij} \leq U$
- Each iteration reduces z by at least 1
- At most mWU iterations
- Many strategies to limit the number of iterations
 - Most negative augmenting cycle at each step
 - Smallest average arc cost
- Expensive algorithm compared to network simplex

Other constraints

- Lower bounds on arcs
- Upper bounds on flow through node
- convex piecewise linear costs on arc flow
- Fixed cost to use a node
- Fixed cost to use an arc
- Proportionality of flow (e.g. $x_{i1} = x_{i2}$)
- Gains and losses of flow along arcs (e.g. power distribution)

Lower bounds on arcs

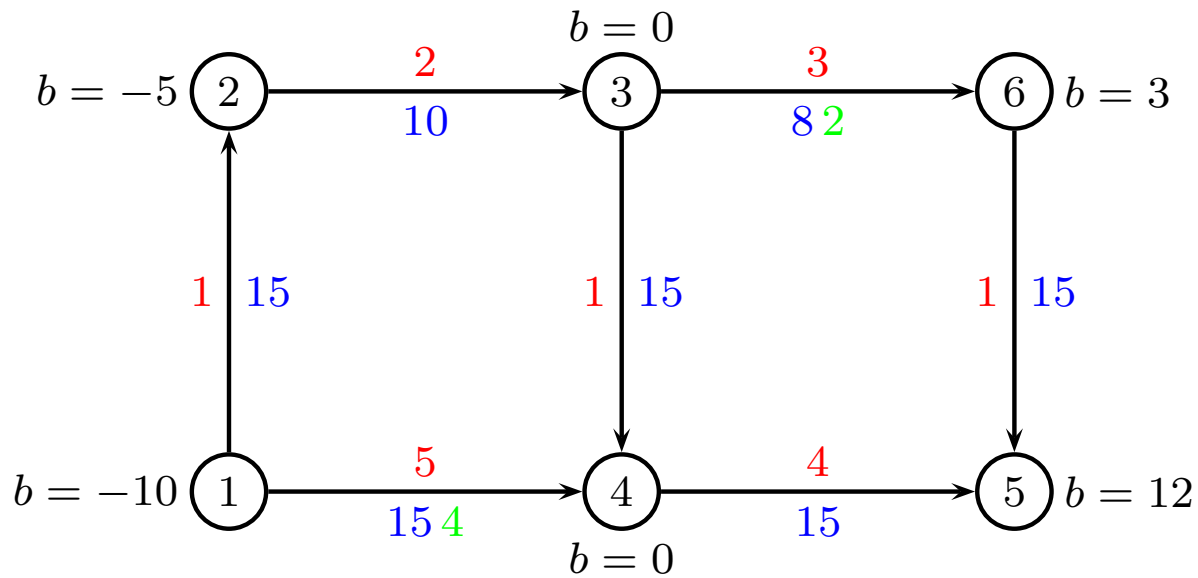
Assume that the flow x_{ij} on arc (i, j) has upper bound u_{ij} and lower bound ℓ_{ij} .

- Replace the upper bound with $u_{ij} := u_{ij} - \ell_{ij}$
- Set the demand at node i to $b_i := b_i + \ell_{ij}$
- Set the demand at node j to $b_j := b_j - \ell_{ij}$

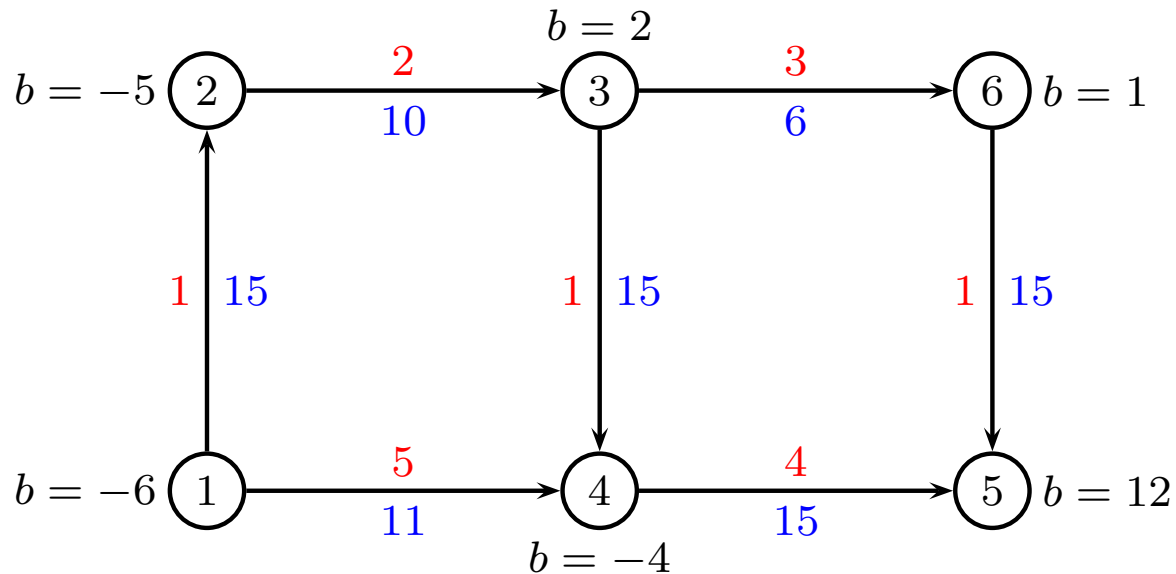
Now, the problem is a minimum cost flow problem with only upper bounds.

After having solved the problem, remember to add ℓ_{ij} to x_{ij}

Lower bounds on arcs



(green color is lower bound). Transformed graph:

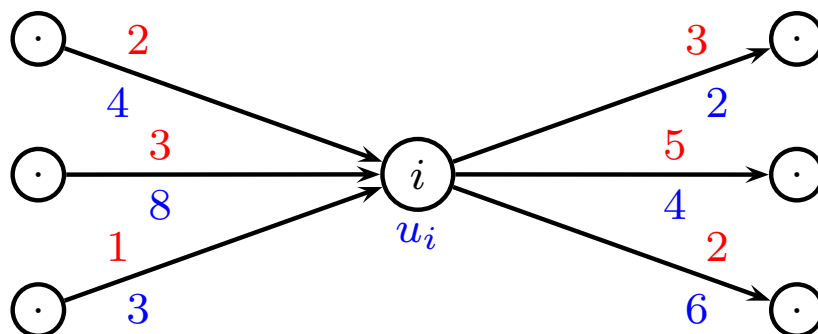


Upper bounds on flow through node

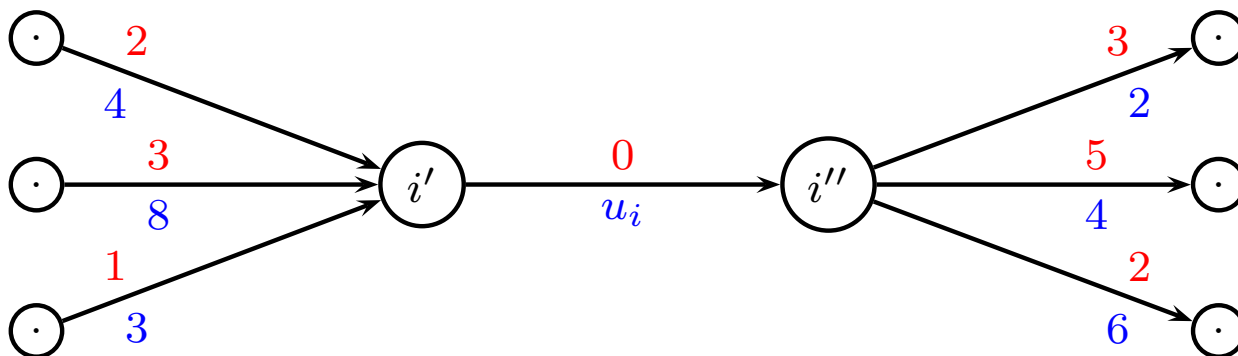
Assume that there is an upper bound u_i on node i

- Replace node i with two nodes i' and i''
- Create an arc from i' to i'' with capacity u_i
- Replace every ingoing arc (j, i) with an arc (j, i')
- Replace every outgoing arc (i, j) with an arc (i'', j)

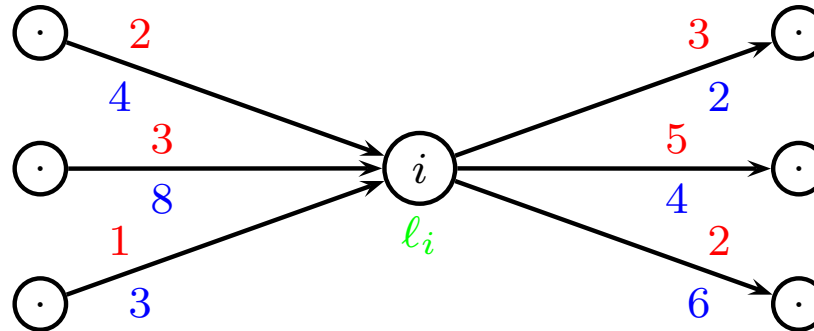
Upper bounds on flow through node



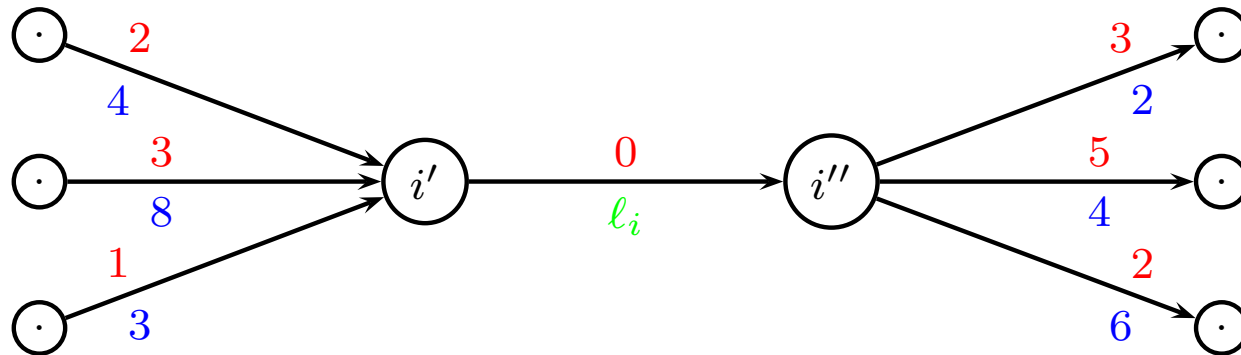
becomes



Lower bounds on flow through node



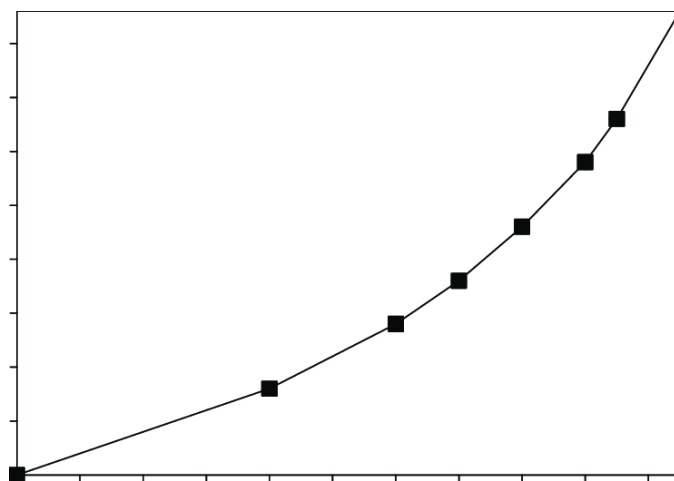
becomes



Convex, piecewise linear costs on arc flows

Assume that the cost on arc (i, j) is a convex piecewise linear function f_{ij} .

- Introduce multiple arcs, corresponding to each portion of the piecewise linear function. Use upper bounds to control the intervals.
- Convexity will assure that cheaper arcs are used before the more expensive arcs



Cannot handle

The following constraints “spoil” the structure of the problem

- Fixed cost to use a node
- Fixed cost to use an arc
- Proportionality of flow (e.g. $x_{i1} = x_{i2}$)
- Gains and losses of flow along arcs (e.g. power distribution)

However, the constraints can be handled using MIP models.
But we do not have constructive algorithms for MIP

Tips-and-tricks

Augmenting circuit method

- check that problem is balanced, otherwise add surplus node to make problem balanced
- When finding an initial solution with max-flow, remember to add source s and sink t
- if you cannot find a negative cycle, use Bellman-Ford
 - if Bellman-Ford continues decreasing distances, you have a negative cycle
 - if Bellman-Ford terminates with stable distances, there are no negative cycles